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Temperature Variation Induced Performance Decline of Perovskite Solar Cells

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temperature, degradation, stability

ABSTRACT

This paper reports on the impact of outdoor temperature variations on the performance of organo metal halide perovskite solar cells (PSCs). It shows that the open-circuit voltage (V_{OC}) of a PSC decreases linearly with increasing temperature. Interestingly, in contrast to these expected trends, the current density (J_{SC}) of PSCs is found to decline strongly below 20% of the initial value upon cycling the temperatures from 10 °C to 60 °C and back. This decline in current density is driven by an increasing series resistance and caused by the fast temperature variations as it is not apparent for solar cells exposed to constant temperatures of the same range. The effect is fully reversible when the devices are kept illuminated at open-circuit for several hours. Given these observations, an explanation that ascribes the temperature variation induced performance decline to ion accumulation at the contacts of the solar cell due to temperature variation induced changes of the built-in field of the PSC is proposed. The effect might be a major obstacle for perovskite photovoltaics, since the devices exposed to real outdoor temperature profiles over 4 hours showed a performance decline > 15% when operated at maximum power point.

INTRODUCTION

Over the course of the past few years, organic-inorganic perovskites have become one of the most promising absorber materials for next generation photovoltaics as well as other optoelectronic applications.¹⁻⁴ They offer a unique combination of key material properties, including long diffusion lengths for electrons and holes, a high absorption coefficient and low material costs.⁵⁻⁷ Starting with a power conversion efficiency (PCE) of 3.8% in 2009 as an absorber in a dye sensitized solar cell, solid state perovskite solar cells (PSC) were first reported

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3 in 2012 with a PCE of 9%.^{3,8} Since then, the progress has led to a world record of 22.7% in
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5 2017.⁹
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8 Driven by the high PCEs, as well as the promise of this material to act as the top device in a
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10 tandem (multijunction) solar cell structure on top a traditional silicon or copper indium gallium
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12 diselenide (CIGS) bottom device, enormous efforts in research are currently directed towards
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14 this technology.^{10,11} Today, the key challenges hindering an industrial breakthrough are: (1)
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16 upscaling the fabrication processes to large areas; (2) the toxicity of solvents and lead, which
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18 remain, to date, essential components of high efficiency PSC fabrication sequences; and (3) the
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20 long-term stability of the material.¹² While solution processing remains the most commonly used
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22 deposition technique, vacuum deposition and inkjet printing are promising alternatives and high-
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24 efficiency PSCs can be fabricated via these techniques.^{13–16} Also, recent reports indicate that less
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26 toxic solvent systems exist that can effectively replace the commonly used harmful solvents like
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28 dimethylformamide.¹⁷ However, so far, the progress on lead-free PSCs that exhibit high PCEs
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30 remains limited.¹⁸ Most importantly, the stability of perovskite photovoltaics remains low, but is
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32 of uttermost importance for the technology's prospects when we consider that crystalline Si and
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34 CIGS photovoltaic modules are among the most durable optoelectronic technologies
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36 commercialized with performance warranties of > 20 years.¹⁹ While this issue is most critical for
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38 the tandem solar cell concept mentioned above, it also remains important for a single-junction
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40 PSC device as the levelized cost of electricity depends greatly on the lifetime of the energy
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42 generator. PSCs have challenges with all major environmental stress factors of a solar module,
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44 namely oxygen, humidity, light and temperature.^{20–25} Although continuous progress on the
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46 stability of perovskite photovoltaics can be achieved via device architecture optimization,
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3 module encapsulation, and variation of the stoichiometry of the perovskite, a sufficient stability
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5 of several decades is still far from reach.
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8 In this work, we investigate the impact of temperature on the performance of PSCs. We study
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10 the temperature-induced performance losses of PSCs by moderate temperature variations, which
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12 impact the charge extraction. Initial studies on the very basic temperature dependence of PSCs
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14 have reported that the PCE is maximal around room temperature with reduced performance
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16 being observed at either higher or lower temperatures.^{26–32} A relative increase in the fill factor
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18 (*FF*) of 2% to 60% over the relevant temperature range of 0 °C to 80 °C is reported from current
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20 density-voltage (*J-V*) measurements.^{29–33} The open-circuit voltage (*V*_{OC}) is shown to decrease
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22 linearly with temperature, which is consistent with the observation of a negative temperature
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24 coefficient of other types of solar cells.^{34,35} With regard to the short-circuit current density (*J*_{SC}),
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26 the existing literature indicates that while it is barely affected by temperature changes below
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28 55°C, it is significantly reduced for higher temperatures.³³ However, there are major
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30 discrepancies between the different findings, arising from differences in solar cell architectures,
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32 statistical variations in cell performance, and measurement protocol.^{20–25} When sweeping the
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34 bias voltage for the determination of the *J-V* characteristics, the resulting device performance
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36 depends on a number of parameters as, e.g., pre-treatment by biasing and illumination, sweeping
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38 direction, scan speed, or delay between voltage change and recording. In addition to temperature
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40 dependent measurements, the effect of temperature cycles on the photovoltaic performance of
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42 PSCs was investigated recently.^{36,37} Cheacharoen *et al.* showed that there is barely any
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44 degradation when the solar cells are encapsulated and cycled between -40 °C and 85 °C while
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46 kept in dark. Unfortunately, this study was not performed under realistic operation conditions i.e.
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48 with a constant illumination and permanent MPP tracking. Domanski *et al.* performed
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temperature cycling measurements under realistic operation conditions with a temperature range between $-10\text{ }^{\circ}\text{C}$ and $65\text{ }^{\circ}\text{C}$ with a ramp of $5\text{ }^{\circ}\text{C}/\text{min}$. However, the solar cells were stressed for more than 40 min at the extreme temperature of $65\text{ }^{\circ}\text{C}$ per cycle which lead to degradation in the hole transport material overshadowing possible effects due to temperature variation. Due to the discrepancies in measurement protocol, a clear understanding of the impact of temperature on performance of PSCs is still missing. Therefore, temperature studies should at least fulfill three criteria: (1) utilize solar cells with a decent efficiency which are stable over several hours under illumination; (2) perform stabilized measurements with zero or minimal changes in bias voltage throughout the complete measurement instead of unreliable sweeps of the J - V -curve; and (3) constantly illuminate the solar cells during the measurements.

In order to understand the impact of temperature variations and cycling on PSCs, we present a detailed and statistically relevant study of the temperature influence on critical parameters of $>15\%$ efficient MAPI based PSCs that are stable under open-circuit conditions for several hours and show a very good stability under MPP conditions (decrease of less than 5% in 4h). We obtain the temperature coefficient for the V_{OC} by applying a temperature cycle to several PSCs. In contrast to previous reports employing JV , the influence of temperature on the V_{OC} is significantly smaller than in conventional solar cell systems. We demonstrate a previously unknown degradation in photocurrent of PSCs which occurs as a result of temperature variations. Due to this degradation, we show that the temperature coefficient of the J_{SC} cannot be measured in this type of PSCs. By investigating the reversibility and the voltage dependence of this degradation, we attribute its origin to ion migration and accumulation at the selective interfaces. Finally, we demonstrate the importance of understanding the underlying current degradation

mechanism while testing PSCs with a realistic outdoor temperature profile. These findings are relevant for industrial application in both single-junction and tandem architectures.

RESULTS AND DISCUSSION

Here, we investigate the temperature coefficients of planar perovskite solar cells based on a methylammonium lead triiodide ($\text{CH}_3\text{NH}_3\text{PbI}_3$) absorber. The layer stack sequence is glass/indium tin oxide (ITO)/titania (TiO_2)/ $\text{CH}_3\text{NH}_3\text{PbI}_3$ /2,2',7,7'-Tetrakis[*N,N*-di(4-methoxyphenyl)amino]-9,9'-spirobifluorene (Spiro-OMeTAD)/gold(Au). Full details regarding device fabrication are given in the experimental section. Our PSCs are well suited for the temperature dependent analyses, since their performance is stable in inert atmosphere when kept for more than ten hours under AM 1.5G illumination (see Figure S1). Furthermore, the solar cells show decent efficiencies, maintaining PCEs of $> 15\%$ throughout these long-term measurements. In order to study the temperature-induced changes in PSCs, we investigate the response of the following solar cell performance parameters with respect to temperature: (1) V_{OC} ; (2) J_{SC} ; and (3) the current at different constant voltages including the maximum point (MPP) voltage. Subsequently, we apply an outdoor temperature profile to the PSCs and monitor their performance.

Open-circuit Voltage. The interrelation of temperature and solar cell parameters is well described for classical semiconductors.³⁸ The parameter most affected in conventional solar cells by an increase in temperature is V_{OC} , which decreases with increasing temperature, predominantly due to a continuous increase in the diode saturation current I_0 with temperature.³⁹ The diode saturation current is mainly affected by the change in the intrinsic carrier concentrations leading to larger rates of recombination with increased temperature.³⁹ According

to the literature, the temperature coefficient related to the decrease in V_{OC} with increasing temperature can be approximated by the following simple linear relation³⁸

$$\beta_{V_{OC}}(T_C) = \frac{10^6}{V_{OC}(25\text{ }^{\circ}\text{C})} \frac{V_{OC}(T_C) - V_{OC}(25\text{ }^{\circ}\text{C})}{T_C - 25\text{ }^{\circ}\text{C}} \quad (1)$$

where $\beta_{V_{OC}}$ is the temperature coefficient in parts per million (ppm)/ $^{\circ}\text{C}$ and T_C is the temperature of the solar cell (for the detailed derivation see Supplementary Material). As depicted in Figure 1a for an exemplary PSC, when increasing the temperature from 10 $^{\circ}\text{C}$ to 60 $^{\circ}\text{C}$, we observe a linear decrease, with an absolute decrease of around 57 mV in V_{OC} . This decrease in V_{OC} is given by a change in the ratio of charge carrier generation rate and recombination rate in the solar cell, which are equal under open-circuit conditions. The charge carrier generation rate is not expected to change notably, as reports indicate that the temperature-induced changes in the bandgap of perovskites are minor^{40,41}, in agreement with observations on other semiconductors used for PV.^{42,43} Hence, in terms of V_{OC} , PSC show the same reduction with increasing temperature as conventional solar cells. With increasing temperature, the carrier density increases which impacts the reverse saturation current by an increase in minority carrier recombination.⁴⁴ Looking in more detail at the V_{OC} evolution with changing temperature in Figure 1a, we find that the V_{OC} takes some minutes to stabilize after changing the temperature stepwise by 5 $^{\circ}\text{C}$ every 7 min. This stabilization is slightly faster if the temperature is decreased compared to increasing temperature (see Figure 1a). Nonetheless, an identification of the stabilized V_{OC} value for each temperature is possible by averaging over the last minute per temperature step (see Figure S2).

We study five PSCs in order to investigate the statistical variation. We find that the $\beta_{V_{oc}}$, calculated as described in Equation (1), of our PSCs is in the range between -900 ppm/°C and -1200 ppm/°C with an average at -1077 ppm/°C, as plotted in Figure 1b. This value is significantly lower than the reported values of around -3000 ppm/°C for crystalline silicon solar cells and -2160 ppm/°C for GaAs cells^{34,35}, indicating that the recombination rates in PSCs are comparably less affected by temperature variations. We note that our finding dissents with the value of -2850 ppm/°C for PSCs based on a fluorine doped tin oxide (FTO)/TiO₂/MAPI/Spiro-OMeTAD/Au stack reported previously by Dunbar *et al.*²⁷ However, our investigation of a

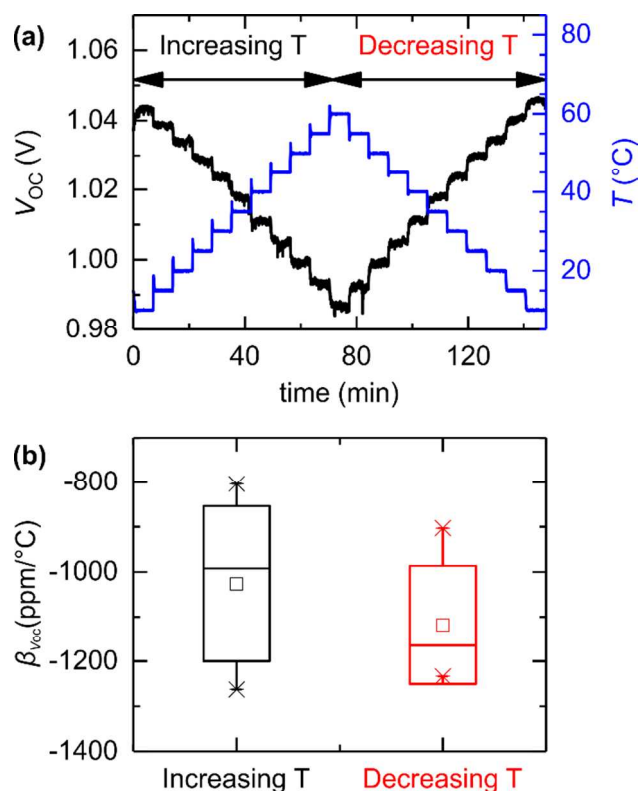


Figure 1. (a) Temperature dependence of V_{oc} as a function of temperature cycling between 10 °C and 60 °C. (b) Temperature coefficient $\beta_{V_{oc}}$ for increasing and decreasing temperature profiles between 10 °C and 60 °C.

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3 statistical relevant amount of solar cells with high PCE gives confidence in our findings for the
4 investigated architecture. We suggest that the discrepancy by a factor of 2.5 between the two
5 results indicates that $\beta_{V_{OC}}$, and therefore the recombination rates, seem to depend strongly either
6 on the architecture or the quality of the PSC. Thus, further research should investigate if there is
7 an impact of PSC parameters such as architecture, perovskite crystal grain size, stability,
8 hysteresis, and performance on $\beta_{V_{OC}}$.
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10
11 **Short-Circuit Current Density.** Following on from V_{OC} , we subsequently evaluate the
12 temperature dependence of J_{SC} . In a similar manner as for V_{OC} , the dependence of J_{SC} for
13 classical semiconductors can be approximated as a linear relation. In general, the J_{SC} in a solar
14 cell is the product of the ideal current and the collection fraction (f_C). The reflects simply the
15 current that could be generated if all incident photons with an energy higher than the bandgap
16 would be converted without any losses, while the collection fraction is a function of the
17 reflection, transmission, parasitic absorption and recombination of charge carriers in the solar
18 cell. We provide a more detailed recapitulation in the supplementary material. For conventional
19 solar cells the changes in J_{SC} with temperature are far smaller compared to the changes in V_{OC} ,
20 meaning that even minor changes in E_g can influence the J_{sc} . Despite the generally accepted
21 linear dependence of J_{SC} on temperature in conventional solar cells, for perovskite solar cells it is
22 yet not clear how exactly the J_{SC} changes with temperature as the bandgap increase causes a
23 reduction in ideal current, while the collection fraction increases typically with temperature.⁴⁰
24 Therefore, depending on which of these aspects dominates, the J_{SC} might either decrease or
25 increase with temperature. Overall, reports indicate that the J_{SC} decreases with temperature for
26 temperatures higher than 25 °C, while there might be an increase or a decrease in J_{SC} between 0
27 °C and 25 °C.^{27–29,31}
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In this study we stress PSCs under two sets of conditions. First PSCs are kept at a range of constant temperatures under short circuit condition. The J_{SC} values of the PSCs stabilize after 80 minutes for all investigated temperatures at values ranging from 90% to 78% of the initial value when permanently kept at short-circuit and constant temperatures, ranging from 15 °C to 55 °C respectively (see **Figure 2a**). Interestingly, we observe a temperature-independent degradation upon short-circuiting the solar cells. Even the solar cell, which is kept at 25 °C before and during the measurement, thus experiences no temperature variation, shows a relative decline in J_{SC} by 11%. Considering the decline at constant room temperature, we cannot explain changes in J_{SC} by changes in bandgap (E_g). Therefore, we conclude that the collection fraction must reduce slowly when short-circuiting the solar cells. We further note that due to the temperature independent decline, conclusions cannot be derived from $J-V$ scans and stabilized measurements are necessary to obtain reliable results.

The smaller difference between the stabilized values of J_{SC} for 25 °C and 40 °C might be explained by an increase in E_g , while the difference between 25 °C and 55 °C in stabilized J_{SC} values is mainly affected by changes in f_c as they are more than 10 times higher than expected from the changes in bandgap.⁴⁵ Hence, not only the ideal current but also the collection fraction seems to decrease in PSCs with temperature. We assume that the stronger reduction in J_{SC} for a constant temperature of 55 °C can be connected to the phase transition which is reported to occur for $\text{CH}_3\text{NH}_3\text{PbI}_3$ at 327-330 K.^{46,47} In contrast to crystalline silicon solar cells, where the J_{SC} increases by 2.4% over the investigated temperature range, we measure a decrease of 13.4% in perovskite solar cells.³⁴

Secondly, in order to obtain a more detailed understanding and determine the temperature coefficient of the J_{SC} as described in literature, we subsequently perform J_{SC} tracking for

temperature cycles with various stabilization times. To our surprise, the J_{SC} under a thermal cycle differs tremendously from the J_{SC} at constant temperatures (Figure 2b). Using the same temperature cycle as for the V_{OC} measurements leads to an enormous reduction in J_{SC} over time. During the first 30 min both constant temperature and temperature cycling show J_{SC} decline of 10% to 20% (see Figure S3 for normalized values). However, afterwards at a temperature of 30 °C, a strong decline in J_{SC} is apparent. At the end of the cycle, the J_{SC} is reduced by 80% of its initial value to a J_{sc} below 4 mA/cm², compared to the maximum degradation by 22% in the case of constant temperature measurement. Moreover, we observe that the J_{SC} also does not recover once the temperature is reduced again.

We observe similar trends for a variety of temperature cycles, for which examples are depicted in Figure S5 and S8. At first glance, one might consider the material degradation as the cause for the low stability of the PSC under temperature stress cycles. However, when comparing to the

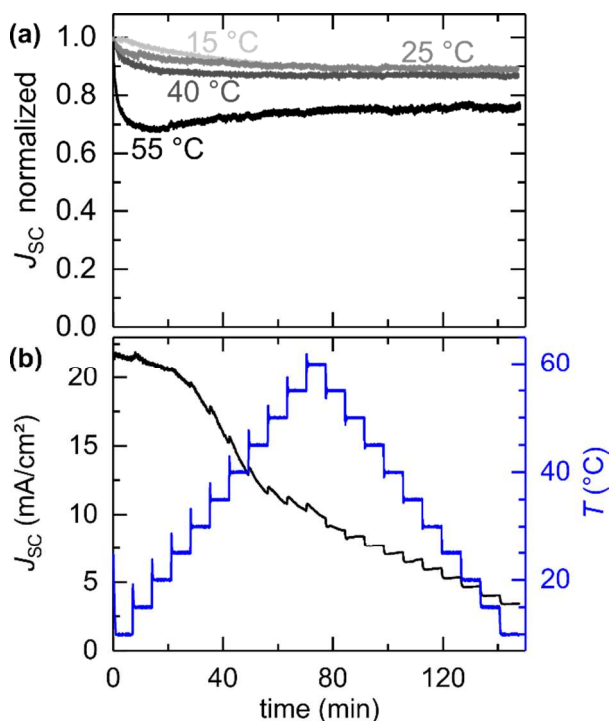


Figure 2. Temperature effect on the short-circuit current of perovskite solar cells. (a) J_{SC} at constant temperatures (b) J_{SC} when temperature cycles are applied.

results of the devices exposed to constant temperature stress (see Figure 2a and Figure S12), the degradation of the temperature cycle is much stronger even though the temperature range is similar. The instant change of temperature itself cannot be the cause, since changing the temperature in a single step from 10 °C to 25 °C leads to a stabilized reduction in J_{SC} that is reached after 50 min (refer Figure S5a). In addition, the degradation is less pronounced but still dominant for smaller temperature ranges (as shown in Figure S5b), increasing with reduced times of the temperature cycling steps.

Comparing all experiments at constant temperature with those at cycled temperature, it becomes apparent that the above reported degradation of J_{SC} is only present in the latter case (see Figure S6). Hence, we identified a degradation mechanism previously unknown in PSCs, which already occurs at temperature variations between 25 °C and 40 °C and cannot be explained with the classical theory on temperature effects in photovoltaics. As a consequence, the extract of a temperature coefficient for the J_{SC} becomes meaningless.

Moreover, it shall be noted that the above presented observation, i.e. strong degradation in J_{SC} upon temperature variations but stabilization of the J_{SC} at constant temperature, is not only limited to solar cells based on TiO_2 as electron transport layer and MAPI as perovskite absorber (see Figure S4). In fact, the qualitatively same degradation in J_{SC} upon temperature variation can be observed also for other solar cell architectures based on SnO_2 as electron transport layer as well as double cation organometal halide perovskites and triple cation organometal halide perovskites. However, the relative degradation seems to be smaller for solar cells with an ETL based on SnO_2 compared to TiO_2 .

Temperature Variation Induced Current Loss at Different Voltages. Having observed very different degradation characteristics of V_{OC} and J_{SC} in PSCs, we study the relevance of these

for the device close to the MPP and other representative voltage conditions in more detail. The key finding (see Figure 3) is a correlation between the continuously applied constant voltage and the current degradation upon temperature cycling. At a reverse voltage of -0.4 V, the degradation is most pronounced with a reduction of the initial current by 43%, while there is less degradation at 0.8 V, where the current is reduced by 3%. On top of the steady reduction in current during temperature cycling, the output current increases/decreases slightly on short timescales as an immediate response to increasing/decreasing temperature steps. Therefore, we conclude that there are two overlapping effects on the collection fraction of the PSC, namely a slow degradation due to constantly changing temperature and a fast response to each individual temperature variation. We note that the first effect outweighs the second one clearly, and that effects on the collection fraction clearly outweigh the changes in bandgap. At 0.8 V, which is

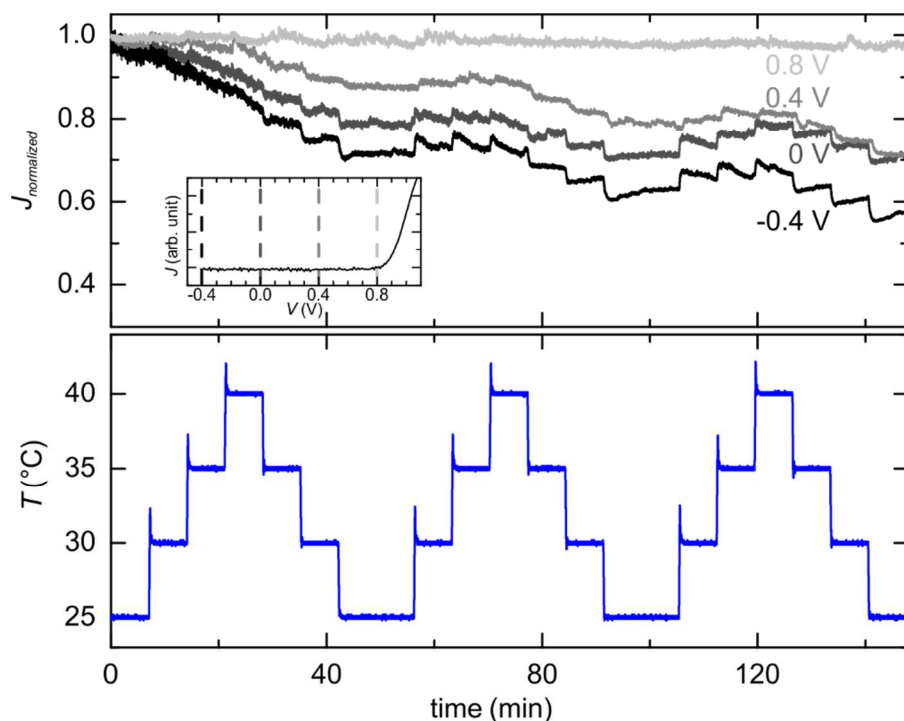


Figure 3. Current density of solar cells stressed with temperature cycles ranging between 25 °C to 40 °C at different voltages, covering a range from close to the MPP to reverse bias.

close to the MPP position, the current is less correlated with the applied temperature. Nonetheless, the degradation at 0.4 V for a very small temperature range indicates that PSCs might be prone to degradation around MPP, because variations in temperature are inevitable in real world applications and possibly even more frequent as well as at higher temperature, which intensifies the degradation (see Figure S5).

Recovery of Temperature Variation Induced Performance Loss. In order to investigate the reversibility of the degradation in current, we light-soak the solar cells at V_{OC} after the constant voltage and temperature cycling measurements and perform J - V scans as a function of time. Even for the PSC that experienced an initial temperature cycling stress at -0.4 V and exhibited the strongest decline in current density, the original solar cell parameters can be restored when the cell is light-soaked at V_{OC} for a period of 600 minutes (see Figure 4a). The full recovery in PCE is also apparent for all other voltages. Thus, we can reverse the degradation in current density upon temperature cycling without any permanent damage of the solar cell. The recovery of FF occurs over a longer timescale (~ 10 min to 600 min), which depends on the previously applied voltage, while the J_{SC} recovers faster (~ 1 min to 70 min), the V_{OC} reaches the initial value almost instantly. As the V_{OC} recovers fastest, we attribute the slow recovery in FF to be correlated with slow changes in internal shunt or series resistance.

We investigate the recovery in further detail for all previously mentioned voltages. The degradation of the PSCs upon temperature cycling is related to an increase in series resistance, which we also find to decrease during recovery, as can be seen in Figure 4a and Figure 4b. Our measurement demonstrates a clearly visible correlation between R_s and the voltage at which the measurement was performed. While the difference in current between the measurement at 0 V

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3 and 0.4 V is not very pronounced, we observe a significant difference in R_S directly after the
4 measurements. As anticipated, there is no change in series resistance for the measurement at
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6 0.8 V. The recovery in series resistance plays the major role in the slow recovery of PCE (see
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8 Figure 4c) in the PSCs. We note that the same recovery does not occur when the solar cells are
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10 stored in dark. Regarding the possible underlying mechanism, given that the outer contacts are
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12 not changed during the measurement, an increase in R_S can only be explained by changes in the
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14 interface between $\text{CH}_3\text{NH}_3\text{PbI}_3$ and the hole or electron selective contacts or inside the
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16 $\text{CH}_3\text{NH}_3\text{PbI}_3$ layer itself.
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21 We note that we can exclude contacting issues as the origin of the increase in series resistance
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23 due to two reason: (1) We only see this effect in PSCs which are stressed with a temperature
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25 cycle. Contacting issues could only arise if the measuring pins could not withstand the whole
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27 temperature range. However, they are specified for a range between -40°C - 80°C . All of our
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29 measurements are well within this range. (2) Another option would be that the contact worsens
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31 with measurement time. We can also exclude this since the series resistance increases during the
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33 temperature stress and is reduced afterwards close to the initial value (see Figure 4b). The cells
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35 are permanently contacted and illuminated from the beginning of the measurements until the end.
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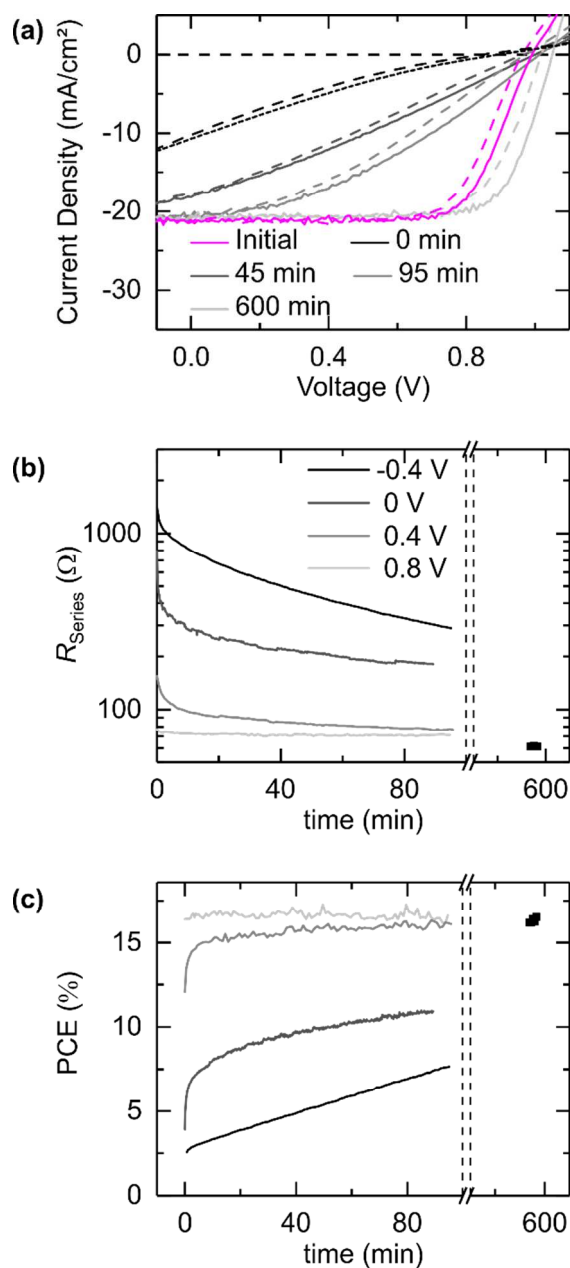


Figure 4. Recovery of the degradation induced by current measurements at constant voltages and temperature cycling. The time refers to the recovery time at V_{OC} after the current measurements. (a) $J-V$ scans of the solar cell kept at -0.4 V. The solar cell recovers to its initial performance when kept at V_{OC} for 600 minutes. (b) Series resistance (R_{Series}) of the PSCs shown in Figure 3. (c) PCE of the same solar cells over time. The recovery was measured for 90 minutes in all devices, the cell kept at -0.4 V previously was additionally light-soaked for 600 minutes total, as indicated by the black points in (b) and (c).

Discussion. Reviewing the results presented above, it becomes apparent that observed phenomena are different from those typically observed in inorganic or organic thin film solar cells. Thus, the origin of the above reported reversible current degradation must relate to specific properties of organometal halide perovskites instead. A prominent and unique property of perovskite solar cells is the apparent ion migration in these devices. Already for the explanation of the short-circuit current of PSCs at constant temperature (see Figure 2a), which has also been investigated by several groups recently, ion migration is typically considered.^{28,48–50} The established understanding is that ion migration on different timescales, depending on the species, leads to an accumulation of ions at the interfaces between the perovskite layer and the hole or electron selective contacts.^{28,48–50} The ions are driven by the built-in field $E_{\text{built-in}}$, which originates from the difference in work functions of the hole transport material (HTM) and the electron transport layer (ETL).^{28,49} According to our temperature dependent measurements on V_{OC} , $E_{\text{built-in}}$ reduces with increasing temperature. Additionally, the ion mobility and thereby the extent of ion migration increases with the temperature of the solar cell.^{51,52} Figure 5a and Figure 5b show on the left-hand side the field in the perovskite solar cells for constant temperatures. As a consequence of the accumulation, the ions induce a polarization field E_{pol} , which is in opposite direction of the built-in field and leads to an effective field E_{eff} that is smaller than $E_{\text{built-in}}$ (see right-hand side of Figure 5a and Figure 5b). As reported elsewhere, the current generation decreases due to the ion accumulation, since the non-radiative recombination at the interfaces is increased, either by an increase in defect states within the bandgap or electrostatic traps.^{53–56}

Considering the strong voltage dependence (see Figure 3), the reversibility (see Figure 4), and long time-scales involved, we propose that the degradation in current density upon temperature cycling is related to a pronounced accumulation of ions at the contacts of the PSCs compared to

the case of constant elevated temperature. In fact, an increase in ion migration due to a temperature gradient in PSCs has been recently reported.⁵⁷ Though, the reported permanent degradation cannot be confirmed in our experiments. A strong increase in ion accumulation at the contacts would lead to the band diagram as presented in Figure 5c with an increase in the space-charge region width and E_{pol} . As a consequence of the ion accumulation the non-radiative recombination at the interface is increased, leading to a decreased current extraction.^{53–56} The difference between the smaller changes in non-radiative recombination with temperature at V_{OC} (see Figure 1) or at 0.8 V (see Figure 3) and the larger changes at J_{sc} (see Figure 2b) are likely due to the different effective fields. The proposed explanation is supported by the fact that a strong dependence of output current-density on the preconditioning and ion distribution in the PSC has been shown in various reports on current-voltage hysteresis.^{48,49,52,53,58–61} Especially, the influence of temperature on hysteresis has been shown.⁶² Furthermore, ion migration depends on the applied electrical field and is fully reversible similar to the reported degradation in J_{sc} upon temperature cycling.^{63,64} Most importantly, it was recently shown that the current-density is also affected on long timescales by ion accumulation (>10 min) in PSCs.⁶³ This might explain our measurements at constant temperature (Figure 2a) and with a single temperature step (Figure S5a), which show that the photocurrent requires > 40 min to stabilize. However, we only use 7 min between two temperature steps in our measurements with temperature cycles. Hence, a steady-state is not reached as the variations of temperature occur significantly faster than the required stabilization time.

It shall be noted that apart from the widely discussed intrinsically apparent ions in organo metal halide perovskites, extrinsic ions such as Li^+ ions of the Spiro-OMeTAD layer might contribute and also migrate towards the TiO_2 layer and influence the electron injection into the

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2
3 ETL.^{48,49} It has been reported that Li^+ ions diffuse towards or away the TiO_2 layer at positive or
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5 negative bias.⁶⁵ This might explain the higher reduction at negative bias in Figure 3. Similarly to
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7 the required stabilization time of several minutes upon varying the temperature (see Figure S5a),
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9 the Li^+ diffusion results in an altered photocurrent upon changing the bias. However, we suspect
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11 that the overall influence of the Li^+ ion diffusion is small compared to the intrinsic ions in the
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13 perovskite absorber, as our solar cells show no strong hysteresis as commonly observed in
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15 devices with strong Li^+ ion discussion.⁶⁵
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19 We acknowledge that future research is needed to unravel the details of the degradation
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21 processes. In particular, it should shed light on the underlying dynamics possible causing the
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23 strong accumulation of ions at the contacts, which seems to be unique to PSCs. Furthermore, it
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25 should be cleared whether extrinsic or intrinsic ions play a role in this degradation. Especially,
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27 the role of the ETL must be investigated since TiO_2 based PSCs are known to have the highest
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29 hysteresis, while solar cells with SnO_2 or C_{60} are less influenced by ion migration.
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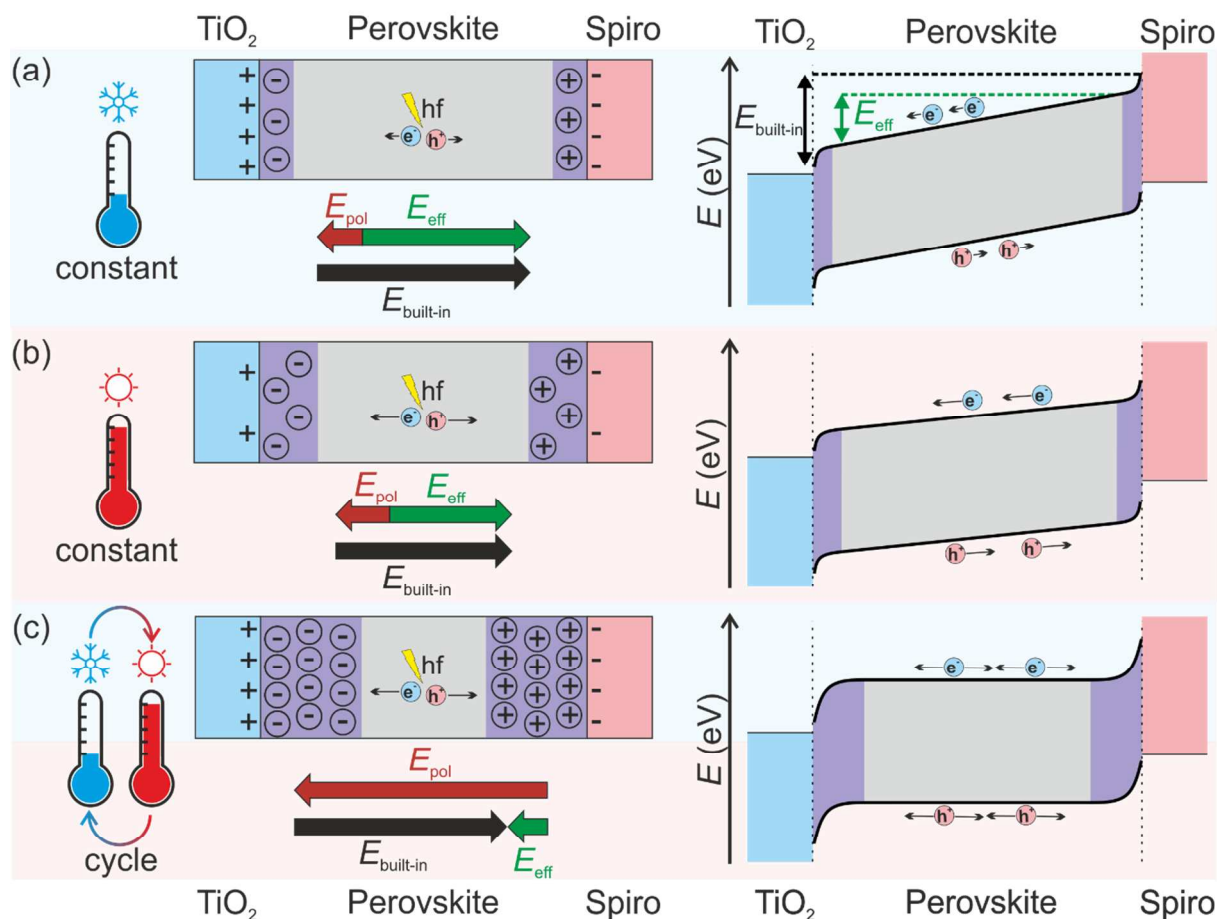


Figure 5. Schematic of the ion migration in an illuminated PSC kept at J_{SC} at different temperatures left and energy band diagrams right. (a) Solar cell at a constant low temperature. The high built-in field supports ion accumulation at the interfaces. (b) Solar cell at constant high temperature. The increased diffusion rate overshadows the reduced built-in field. The space charge region is slightly increased (c) Schematic of the situation in a system with temperature cycles. The disadvantages of a high built-in field and high diffusion rates are combined. Therefore, more ions accumulate at the interfaces, leading to a significantly reduced effective field and a large space charge region.

Outdoor Temperature Profile Induced Performance Decline. The above reported temperature variation induced decline of current generation in PSC will also have an impact on the outdoor performance of this technology. In order to illustrate this impact, we apply a previously recorded real outdoor temperature profile (see Figure 6a) of a perovskite module to our PSCs while illuminating them with an air-mass (AM) 1.5 global spectrum under the solar simulator in nitrogen atmosphere. Hence, with the replicated temperature profile, the PSCs experience no other stress factor e.g. humidity which would be present in an outdoor test. We note, that unlike in our PSC, the recorded temperature changes in the perovskite module are mostly due to variations in illumination intensity and not due to changes in ambient temperature. The temperatures and the temperature steps with up to 2.5 °C/min of the outdoor profile are high compared to the cycles in Section 2.3. Furthermore, the changes in temperature are more frequent with one minute between two temperature steps. Figure 6b shows the V_{OC} in dependence of the temperature profile, which varies between 35 °C and 55 °C. As observed above, the V_{OC} reacts quickly to temperature changes and declines with temperature. We observe a degradation of around 50 mV, which means a relative change of 5%. In order to determine the impact of the outdoor temperature profile on the PCE, we set the sampling voltage at the maximum power point voltage (V_{MPP}) and tracked the current density. As shown in Figure 6c, the PCE decreases significantly during the temperature profile. The total decline is 15% relative, which is much more than the decrease observed in V_{OC} (Figure 6a). Thus, we conclude that most of the performance loss due to exposure to outdoor like temperature conditions can be explained with ion accumulation. Overall, our study demonstrates that individual temperature steps have little impact on the performance of PSCs, but the cycling of temperature and variation

in temperature as apparent in real outdoor temperature profiles lead to a strong degradation in current generation. The origin of this degradation mechanism needs to be identified as it is critical for the industrial applicability of perovskite photovoltaics. Future studies will be directed to investigate whether other architectures of perovskite solar cells or other organo-metal halide perovskites are more robust to this effect. Possible device architecture where the ion accumulation and charge carrier recombination at the HTM/CH₃NH₃PbI₃ and ETL/CH₃NH₃PbI₃ interfaces can be omitted at V_{MPP} might prevent a continuous degradation of the solar cell under outdoor temperature cycling. In general, a high FF is preferable, as this leads to a higher V_{MPP} and low $E_{built-in}$, conditions for which we demonstrated that temperature variation induced declines in current density are less pronounced (see Figure 3). Especially two-terminal tandem architectures are expected to suffer from the reported current degradation upon temperature variation.

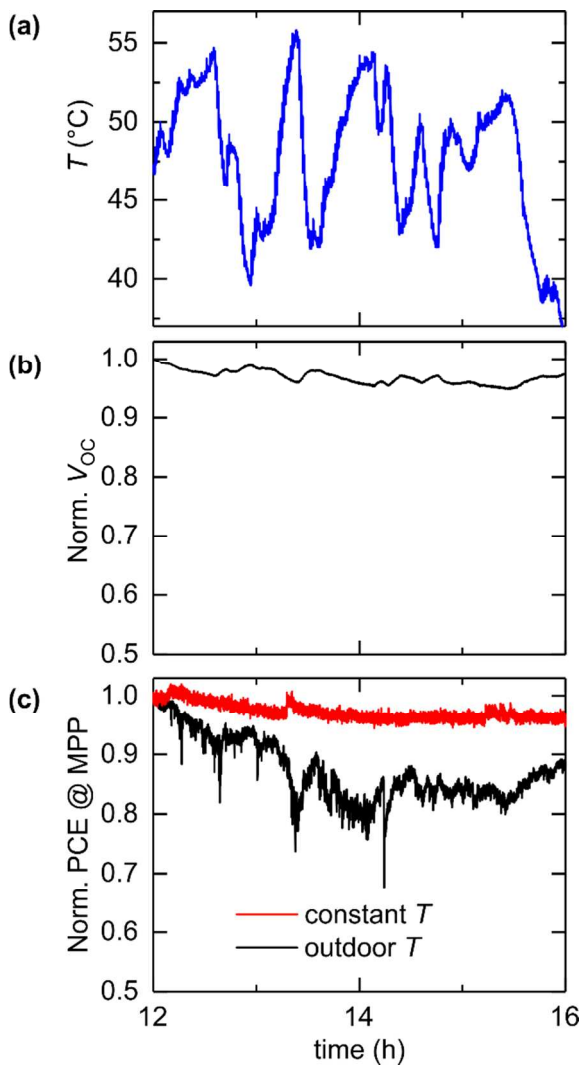


Figure 6. Measurements on perovskite solar cells with an outdoor temperature profile. (a) Shows the replicated temperature profile. (b) and (c) show the open-circuit voltage and maximum power point measurement of a solar cell stressed with this temperature profile. The constant temperature measurement depicted in (c) is recorded at 25 °C.

CONCLUSIONS

In this work, we investigate the degradation mechanisms in PSCs upon temperature variations, which are inevitable in real-world applications. By introducing a statistical approach under

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3 stabilized operating conditions, we show that the V_{OC} of PSCs follows a well-known moderate
4 linear decrease with increasing temperature and is less affected by temperature than the V_{OC} in
5 conventional solar cells. In contrast to the V_{OC} , the effect of temperature on the J_{SC} differs from
6 classical solar cell theory. We demonstrate a previously unknown degradation in current density
7 upon temperature cycling that depends on the applied voltage and is more pronounced for
8 reverse bias than for voltages close to open-circuit. This degradation is only apparent when the
9 temperature is varied during the operation of the PSCs as the J_{SC} stabilizes with constant elevated
10 temperatures. Moreover, we show that the decline in current density for different voltages is
11 driven by an increasing series resistance and can be reversed by illuminating the solar cells at
12 open-circuit for several hours. Combining these observations, we propose that ion accumulation
13 at the interfaces of the selective contacts is promoted by continuous variations in temperature
14 leading to the increase in series resistance and current reduction. By applying an outdoor
15 temperature profile to perovskite solar cells at MPP and V_{OC} conditions, we demonstrate the
16 possible implications of this effect for application. Our results indicate that having a high V_{MPP}
17 might prevent the accumulation of ions, reducing the sensitivity of perovskite solar cells to
18 temperature variations and thereby overcoming one of the obstacles for outdoor operation. We
19 consider further investigations to unravel and to avoid this effect a mandatory prerequisite for
20 outdoor applications of perovskite solar cells.
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47 EXPERIMENTAL SECTION

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49 **Substrate preparation.** The glass substrates with patterned indium tin oxide (ITO) electrodes
50 were purchased from Luminescence Technology. The ITO coated glass was cleaned with
51 ultrasonic baths of detergent, deionized water, acetone, and iso-propanol each for 10 minutes.
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Electron transport layer preparation.

The TiO₂ layer was fabricated as described elsewhere.⁶⁶ Titanium dioxide (TiO₂) pellets purchased from Prof. Feierabend GmbH were evaporated at a rate of 1 Å/s onto ITO substrates in an Angstrom Engineering evaporation system, equipped with an electron beam source. A partial oxygen (O₂) pressure of 1.7×10^{-4} Torr was used during the deposition, to maintain the stoichiometry of the film. The final TiO₂ layer thickness is 20 nm.

The SnO₂ layer was fabricated by further diluting a 15% aqueous colloidal dispersion of SnO₂ (Alfa Aesar) to a final concentration of 2.04%. The dispersion was spin-coated at a speed of 4000 rpm for 30 s on the prepared ITO substrates followed by an annealing step at 250 °C for 30 min in air.

Perovskite layer preparation.

For MAPI films, Pb(CH₃CO₂)₂·3H₂O, PbCl₂, and CH₃NH₃I were dissolved in *N,N*-dimethylformamide (DMF). The concentration of lead source, Pb(CH₃CO₂)₂·3H₂O together with PbCl₂, was 0.8 M, while the concentration of CH₃NH₃I was 2.4 M (Pb : CH₃NH₃I = 1:3). The PbCl₂ molar fraction in the lead source was 20%. The precursor solution was stirred at room temperature for 10 min, and then spin coated at 3000 rpm for 60 s onto the TiO₂ layer. The obtained films were annealed on a hot plate at 130 °C for 10 min to form a perovskite crystal structure.

For double cations, PbI₂ (1.1 M), PbBr₂ (0.18 M), formamidinium iodide (1M) and methylammonium bromide (0.18 M) were dissolved in a DMF:DMSO mixture of 4:1 (v:v). The solution was spin coated in a two steps program beginning with 1000 rpm for 10 s followed by a

step at 5000 rpm for 20 s. 100 μ l of chlorobenzene was poured on the substrate. Subsequently, the perovskite layers were annealed at 100 $^{\circ}$ C for one hour.

For triple cation films, PbI_2 (1.1 M), PbBr_2 (0.2 M), formamidinium iodide (1M) and methylammonium bromide (0.2 M) were dissolved in a DMF:DMSO mixture of 4:1 (v:v) to form a double cation solution. Additionally, 88.9 μ l of a 1.5 M CsI stock solution in DMSO was added per ml of double cation solution to receive a 10% cation ratio of Cs. The triple cation solution was spin coated in a two steps program beginning with 1000 rpm for 10 s followed by a fast step at 6000 rpm for 20 s. Seven seconds before the end of the second step, 100 μ l of chlorobenzene was poured on the substrate. Subsequently, the perovskite layers were annealed at 100 $^{\circ}$ C for one hour.

Further solar cell preparation.

Subsequently, 80 mg/ml Spiro-OMeTAD solution doped with 17.5 μ l lithium bis(trifluoromethanesulfonyl) imide (520 mg/ml in acetonitrile) and 28.5 μ l 4-tert-butylpyridine was spin-coated onto the perovskite films. All of the spin-coating processes mentioned above were performed in a nitrogen filled glove box. The perovskite films coated with Spiro-OMeTAD were then exposed to air overnight for Spiro-OMeTAD oxygen doping. The small area devices were completed by depositing a 60 nm Au layer onto the Spiro-OMeTAD through shadow masks, defining an active area of 0.105 cm^2 . Figure S13 shows a cross section SEM for the glass/ITO/ TiO_2 /MAPI/Spiro/Au configuration.

Characterization. Solar device characteristics are measured in a Newport solar simulator (xenon lamp) under AM1.5G conditions in both forward and backward direction with fixed scan rates of about 1 V/s with a Keithley 2400 sourcemeter. The temperature of the devices is actively controlled by a Peltier element connected to a microcontroller. A schematic illustration of the

setup is provided in the supplementary information. All temperature dependent measurements were performed on individual solar cells to avoid influences of the temperature history on the outcome of the measurement.

Outdoor measurement. The measurements of the outdoor temperature profile have been performed by the ENGIE Laborelec Solar lab on May 15th 2016. A 225 cm² glass-glass packaged perovskite module has been mounted with a temperature sensor attached to the rear electrode on an outdoor testbench oriented South with 35° inclination. The temperature has been recorded with timesteps of one minute. The recorded temperature profile was replicated by our temperature controlled measurement setup to simulate the outdoor temperature conditions under inert atmosphere. Figure S9 shows a comparison between the measured outdoor profile and the replicated temperature profile.

ASSOCIATED CONTENT

Supporting Information. Theoretical Recapitulation of Temperature Effects in Solar Cells; Long term stability of the investigated PSCs; Individual open-circuit voltage measurements and linear fits; Normalized J_{SC} of Figure 2b; Degradation on other PSC architectures; Short circuit current dependence of perovskite solar cells on the temperature dynamics; Statistics of short-circuit current degradation; Recovery of J_{SC} , FF and V_{OC} ; Reproducibility of current degradation; Comparison between measured and replicated outdoor temperature profile; Scheme of the temperature controlled measurement setup; J - V -characteristics at different temperatures; Comparison between constant and varying temperature; Cross section SEM

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Author Contributions

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TOC FIGURE

